Where Should the Gradient Go?

When we fine-tune a small VLM on a limited budget, should we adapt the vision side or the language side?

CS 8803-VLM, Fall 2026 ~ Zsolt Kira

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Outline

01	**Problem Statement**	Which module should receive a fixed LoRA budget?
02	**Motivation**	Small models make gradient placement consequential
03	**Prior work**	Diagnostics and recipes exist, but the link is missing
04	**Study design**	Measure the bottleneck, test placement, evaluate prediction
05	**Timeline**	Midterm go/no-go criteria and final transfer test

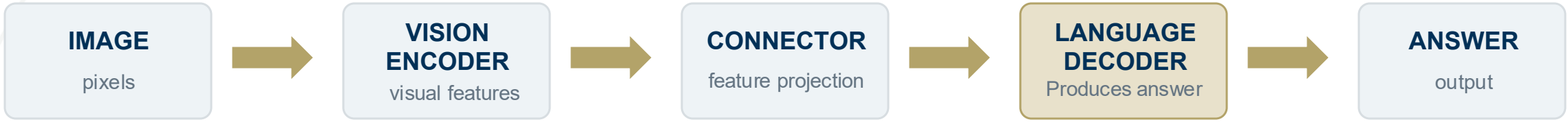
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Problem statement

A pre-training diagnostic should determine where limited adaptation capacity goes

At its core, this is a decision
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Given a fixed LoRA budget B, **which part of a small VLM should receive the gradient?** Our language-side baseline freezes the entire visual pathway and applies LoRA to the decoder.



Inputs A small pretrained VLM, training data, a separate diagnostic set, and a fixed adaptation budget
Outputs A measured error profile, a placement prediction, and checkpoints from both placements
Success means Lower selection regret than always committing to one placement
with uncertainty reported

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Motivation

Why does this matter? Because the budget is small and indivisible: spend it on the wrong module, and it can't repair the actual failure.

Target : Compact deployment

On-device, edge, cost-constrained products and single-GPU labs

Fixed adaptation budget

A few GPU-hours and a limited number of trainable parameters

Placement matters

That same budget can target perception, alignment, or reasoning

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FAILURE EXAMPLE

Chart value

$48M

VLM reads

$43M

Better arithmetic cannot recover an answer built from the wrong visual fact: tuning text reasoning is totally wasted !

Who benefits	Chart, document and scientific-figure systems Few-shot industrial or medical deployments, plus robotics and VLA adapters

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Related work and the gap

We test whether oracle-text rescue predicts a choice between visual and language adaptation

Research stream	What is known	Our question
Perception diagnostics	Changing the visual information available in text can reveal sensitivity to visual input. MathVerse; CapGeo-Bench	Does oracle-text rescue predict which adaptation placement works better?
LoRA placement	PLoP selects adapter module types; Image-LoRA selects attention heads and adapts visual-token positions.	Can an inference-only diagnostic guide a choice between the visual pathway and the language decoder?
Visual fine-tuning	Several visual fine-tuning methods improve average performance, but gains vary across tasks. CoVFT	Does the diagnostic track the vision-minus-language accuracy difference across task-model pairs?

Contribution: a controlled test of whether oracle-text rescue predicts adaptation placement

Sources: MathVerse Zhang et al. (2024); CapGeo-Bench Li et al. (2025); PLoP Hayou et al. (2025); Image-LoRA Luo et al. (2025); CoVFT Zhou et al. (2026)

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Proposed approach

Record placement predictions before training, then test them with matched-budget comparisons

Stage A · Diagnostic

Image + question
deployed condition

Oracle text + question
substitutes audited visual facts

Question only
measures nonvisual answerability

Error profile

Oracle-rescued errors
Residual errors
Question-only correct

p = oracle-rescued fraction among image errors not answered correctly with the question alone

Stage B · Placement grid

Two placement

- Language decoder LoRA
- Vision encoder + connector LoRA

Shared training objective

- Answer-only SFT
- Oracle text used only for diagnosis

Trainable parameters matched within 10%
Same training data and optimizer steps
Exactly 3 seeds per placement
6 runs per pair; 24 runs total

Stage C · Decision rule

Candidate decision rule

- **p > 0.5: choose vision-side adaptation**
- **Otherwise: choose language-side adaptation**

Fix the rule before observing training outcomes

Primary tests

H1: Higher oracle rescue predicts a larger vision-minus-language accuracy difference
Compare selection regret with always-language and always-vision

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Timeline and evaluation checkpoints

The midterm reports diagnostic variation and progress on the full three-seed comparison

W5 · Sep 22–26	W6 · Sep 29–Oct 3	W7 · Oct 6–10	W8 · Oct 13–15
Build and audit oracle text Check 100 examples per task Finalize splits and evaluation	Profiles for both 2B models ChartQA + Geometry3K Record all four predictions	Launch the 24-run grid Two placements per pair Three seeds per placement	Report completed comparisons Show all available seed results Mark incomplete sets as preliminary

MIDTERM CHECKPOINTS

1 · Diagnostic quality

Are oracle descriptions faithful, and do rescue fractions vary across the four pairs?

2 · Effect uncertainty

How large are the placement differences relative to seed variability? Keep unresolved results inconclusive.

Optional after core runs Evaluate chart-adapted checkpoints on CharXiv and ChartQAPro without additional fine-tuning

Final phase Complete all 24 runs; report prediction outcomes, selection regret, uncertainty, and the code and audit packet

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Thank-you

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